Terms Practice: Distance, Displacement and Basic Motion Name: Gene P.2

Key Terms: Displacement, Distance, Vector, Scalar, Magnitude, Direction

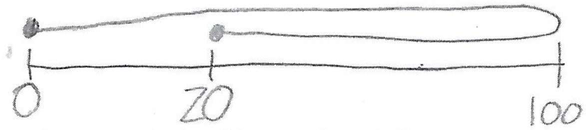
8/21/26

Underline Magnitudes, Circle directions, Label as V in the blank (if it's a Vector)

10 meters ____	+22 cm/s __V__	70 mph ____
55 inches forward __V__	-15 cm/s __V__	70 mph East __V__
8 miles North __V__	4 inches ____	2 m backward __V__

Scenario: *A runner moves from the start line (initial position = 0 m) to the finish line (position = 100 m) in 12 seconds. He then stumbles back to position 20 m taking an additional 48 seconds. Sketch a number line with essential information labeled.*

Sketch →



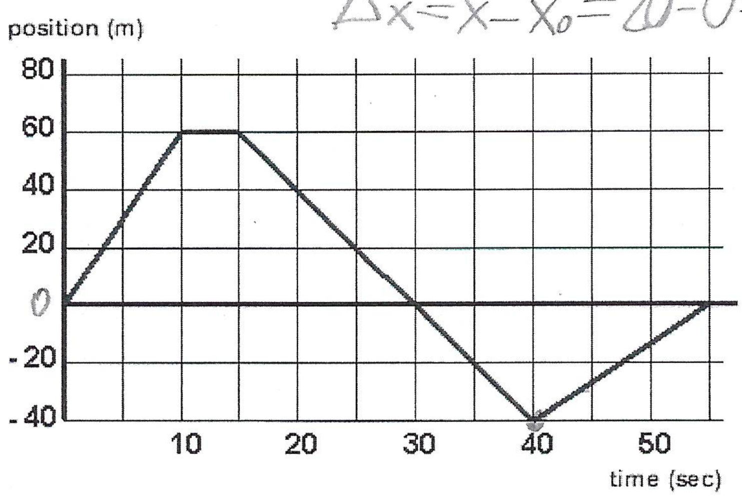
1. What is the distance (a scalar quantity) of the entire trip?

180m

2. What is the displacement (a vector quantity) of the entire trip?

A. Magnitude (# & units)? B. Direction?

$\Delta x = x - x_0 = 20 - 0 = 20m$ fwd



For #6-10, use the graph above (Don't forget a vector's magnitude & direction). Moving in positive direction is considered forward (**fwd**). Moving in the negative direction is considered backward (**bwd).**

3. What is the ***distance traveled*** during t = 0 to t = 30 sec? $60 + 60 = 120m$

4. What is the displacement during t = 0 to t = 30 sec? $0 - 0 = 0m$ fwd

$x - x_0$

5. What is the displacement during t = 15 sec to t = 40 sec? $-40 - 60 = -100m$ (bwd)

6. What is the displacement during t = 40 sec to t = 55 sec? $0 - (-40) = 40m$ (fwd)

7. What is the displacement during from t = 0 to t = 55 sec? $0 - 0 = 0$

8. What is the ***distance traveled*** for the whole trip? (0-55 sec)

$60 + 60 + 40 + 40 = 200m$

Average Velocity – Problem Solving Practice **v = d/t** **Name:**

List knowns and unknowns with variables attached. Substitute and solve. Answer bank at end of problems.

Unit check: Distance ______, Displacement ______, Time _______, Speed or Velocity ______

1. A SeaKing running back went from the 10 yard line to the 30 yard line in 5 seconds. What was his average velocity (yards/sec)?

2. On the next play, a running back went from the 30 yard line to the 45 yard line. At the 45 yard line he was hit and pushed backward 3 yards. The entire play (forward and backward) occurred in a 4 second time interval.
   a. What was his displacement for the whole motion?(fwd & bwd together)

   $\Delta x = x - x_0 = 42 - 30 = 12$ yds

   b. What was his average velocity for the whole motion if 4 seconds elapsed? 30 35 40 45

   $V = \frac{\Delta x}{t} = \frac{12 \text{ yds}}{4 \text{ sec}} = 3$ yd/s $\quad x_0 = 30 \quad x = 42$

3. A student walks at 3 m/s toward their next class. How long will it take them to reach the class if it is 150 meters away?

   $T = \frac{\Delta x}{V} = \frac{150}{3 \text{ m/s}} = 50 \text{ sec}$

4. A bus moving at 20 meters/second in a straight line goes for 30 minutes before it stops. How far did it travel? (careful w/ units!)

5. A runner moving at 5.0 m/s North ran 1600 meters. How much time elapsed? (in seconds, in minutes)

   $t = \frac{\Delta x}{V} = \frac{1600}{5} = 320 \text{ sec} \qquad \frac{320 \text{ sec}}{} \cdot \frac{1 \text{ min}}{60 \text{ sec}} = 5.33 \text{ min}$

6. A delivery truck drives for 2 hours at 50 miles per hour North and then 3 hours at 30 miles per hour North. What was the magnitude and direction of the average velocity of the delivery truck?

   $\Delta x_1 = Vt = (50)(2) = 100 \text{ mi}$
   $\Delta x_2 = Vt = 30(3) = 90 \text{ mi}$

   $\bar{V} = \frac{\Delta x_{total}}{t} = \frac{100+90}{2+3} = 38$ mph North

7. Rafael Nadal can hit a serve at 130 miles per hour (57.8 meters/sec). Find out how much time it takes the serve to reach the opponent who is 78 ft away (23.6 meters).

8. If you walked 200 meters in 1 minute, 400 meters in 4 minutes, and 800 meters in 10 minutes, find your average velocity for the whole walk. (meters/minute)

9. America's Cup Sailing Final Race 19. Boat NZ (Team Emirates New Zealand) crosses the starting line first moving at 40.0 knots (20.4 meters/second) while Boat USA (Team Oracle USA) crosses the line 5 seconds later at 41.0 knots (20.95 meters/second). If the first mark (turn) is 960 meters away, which boat gets to the turn first? (Challenge: What is their lead in meters?)

10. It is approximately a 500 mile drive to San Francisco. If you spend the first 150 miles going 40 mph due to traffic congestion, how fast will you need to go for the rest of the trip in order to have an average velocity of 60 mph for the whole trip?

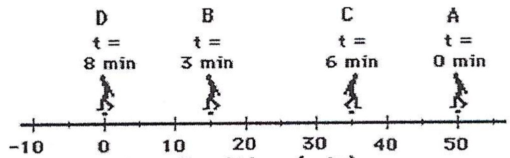
Answer Bank (Cross off answers below as you use them)

0.41	3	4	5.33	12	38	50	57.8	76	93	191	320	36000

Which of the following statements about distance and/or displacement are TRUE? List all that apply.

1. A person makes a round-trip journey, finishing where she started. The displacement for the trip is 0 and the distance is some nonzero value. T
2. The phrase "20 m, west" likely describes the displacement for a motion. T
3. The diagram below depicts the path of a person walking to and fro from position A to B to C to D. The distance for this motion is 100 yds. 35+20+35 = 90 F
4. For the same diagram below, the displacement is 50 yds.

back —> vector (# of direction)



Which of the following statements about velocity, speed & acceleration are TRUE? List all that apply.

1. Velocity is a vector quantity and speed is a scalar quantity. T
   (# of direction) (# only)
2. Both speed and velocity refer to how fast an object is moving. T
3. Person X moves from location A to location B in 5 seconds. Person Y moves between the same two locations in 10 seconds. Person Y is moving with twice the speed as person X. F
4. The velocity of an object refers to the rate at which the object's position changes. T
5. The phrase "30 mi/hr, west" likely refers to a scalar quantity. F
6. An object which is slowing down has an acceleration.
7. Acceleration is the rate at which the velocity changes.
8. An object that is accelerating is moving fast.

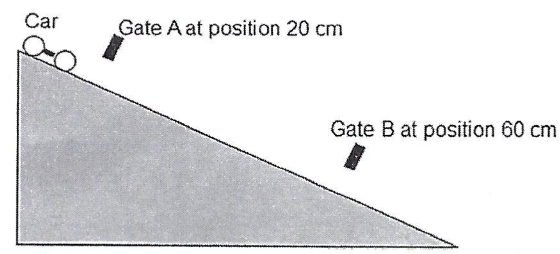
Study Guide for Motion Quiz Name: Gene

Terms (Be able to define the following) (Include units, and if it is a vector or not) Reference point/origin, Position, Displacement, Distance, Speed, Vector vs. Scalar, Average Velocity, Instantaneous Velocity, Acceleration.

Car & Ramp Labs

Width of wing on car = 1 cm (popsicle stick)

Time in Gate A = 0.0148 s
Time in Gate B = 0.0058 s
Time from A-B = 0.3496 s



1. Find the Displacement AB 40cm
   (Show the numbers you use before you solve)
2. Find the Average Velocity A to B (V_{AB}) = $\frac{40}{0.3496} = 114$
3. Find the Instantaneous Velocity (V_A) = $1/.0148 = 67.6$
4. Find the Instantaneous Velocity (V_B) = $\frac{1}{.0058} = 172$
5. Find the acceleration down the ramp =

Lab Answer Check **40** **67.6** **114** **172** **299**
(cm, cm/s, cm/s/s)

Mixed Problem Solving Practice (List knowns and unknowns, Show Equation, Substitute & Solve!) See Equations in notes….